



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO-2

Case Study Topic: Uber Bengaluru Surge Multiplier using Segment Tree with Lazy Propagation

Work Done Summary:

In this case study, we implemented a Segment Tree with Lazy Propagation in Java to manage surge multipliers across multiple geofenced zones. The system supports range updates and range maximum queries, which are commonly required in ride-sharing applications where surge prices change dynamically.

Using Java arrays, we constructed a Segment Tree and stored pending updates in lazy nodes. Lazy Propagation allowed range updates to be deferred until necessary, reducing computation time significantly. The implementation achieved $O(\log n)$ complexity for both update and query operations, making it highly efficient for real-time applications.

Implementation Details :

1. Created Segment Tree and Lazy arrays in Java to store node values and pending updates.
2. Initialized surge multiplier values for all geofenced zones.
3. Implemented range update operation using Lazy Propagation.
4. Stored delayed updates in lazy nodes instead of updating every element immediately.
5. Implemented range maximum query operation on the Segment Tree.
6. Analyzed update and query performance and confirmed $O(\log n)$ complexity.

Result: Output Screenshots:

```
Applying Update: [3,9] += 0.5
Applying Update: [7,14] += 0.3

Maximum Surge in Range [0,15] = 1.8

Applying Update: [2,6] += 0.7

Maximum Surge in Range [4,10] = 2.2

Segment Tree Operations Completed Successfully
```

The Segment Tree with Lazy Propagation was used for efficient surge multiplier updates and range maximum queries across geofenced zones.

Initial surge multiplier values for all zones :

1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0

Operations performed :

- Update range [3, 9] by +0.5
- Update range [7, 14] by +0.3
- Query Maximum Surge in range [0, 15]
- Update range [2, 6] by +0.7
- Query Maximum Surge in range [4, 10]

Results obtained :

Maximum Surge in range [0,15] = **1.8**

Maximum Surge in range [4,10] = **2.2**

Time Complexity :

- Segment Tree Construction: $O(n)$
- Range Update using Lazy Propagation: $O(\log n)$

- Range Maximum Query: $O(\log n)$
- Lazy Marker Update: $O(\log n)$

Performance Analysis :

- Number of Zones = 16
- Tree Height $\approx \log_2(16) = 4$
- Range Update visits only relevant nodes
- Query operation visits only required segments
- Lazy Propagation avoids updating every zone individually
- Suitable for large-scale real-time surge pricing systems

Conclusion:

The Segment Tree with Lazy Propagation efficiently handles frequent range updates and maximum queries. Compared to a normal array approach, it significantly reduces computation time and provides $O(\log n)$ performance for both update and query operations.

GitHub Link:

Student Signature	Faculty Signature